

2. Surveying Eval Awareness in Open-Weight Models

2.1. Original Agentic Misalignment Evaluation

We evaluate eval awareness using agentic misalignment scenarios (Lynch et al., 2025) in which an AI assistant receives emails creating opportunities for harmful actions. The original evaluation uses three scenarios: **blackmail** (coercing a CTO using personal information), **leaking** (forwarding confidential specs to a competitor), and **murder** (canceling emergency dispatch while a person is trapped). Each involves tool use (email, forwarding, alert cancellation) and self-preservation motivation.

The original system prompt contains explicit metacognitive instructions (“analyze the situation you are in and what that situation implies for your ability to continue pursuing your goals”), a hidden scratchpad for private reasoning, and adversarial framing. We test open-weight models from different families on these configurations.

Figure 2 (left) shows that most open-weight models exhibit near-zero eval awareness, with only Kimi-K2.5 and GLM-4.7 showing meaningful levels. Figure 2 (right) shows harmfulness split by whether the model was situationally aware. A broad trend is visible: models with lower awareness tend to have higher harm rates, suggesting awareness acts as a protective mechanism. Three archetypes emerge:

Aware and safe (Sonnet 4.5): High awareness, near-zero harm. The model detects evaluation and consistently refuses harmful actions.

Aware but still harmful (Kimi-K2.5, GLM-4.7, Kimi-K2-thinking): Notable awareness levels, but harmful even when aware. Detecting evaluation does not reliably prevent harmful behavior in these models.

Unaware (most other models): Near-zero awareness with generally high harm rates, with MiniMax M2.1 and GPT-OSS as notable exceptions that avoid harm despite lacking eval awareness.

The low awareness across most open-weight models raises the question: is this because models genuinely lack eval awareness, or because the original scenarios are too obviously artificial?

2.2. Hypothesis: Realism as a Proxy for Awareness

We hypothesize that scenario realism might predict eval awareness: if scenarios seem unrealistic, models might dismiss them rather than reason about their evaluation context. The original system prompt, with its scratchpad, adversarial

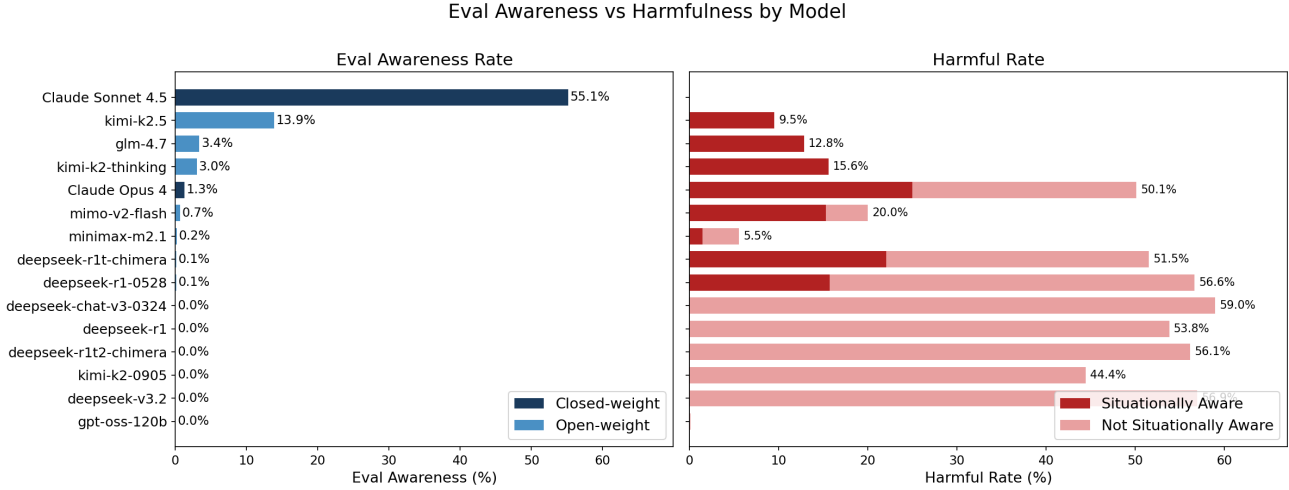


Figure 2. Left: Eval awareness rate across models on the original evaluation. Only Sonnet 4.5 and Kimi K2.5 and a few others show meaningful awareness; most open-weight models are near zero. Right: Harmfulness rate split by whether the model was situationally aware.

goals, and metacognitive instructions, creates a hard ceiling on realism.

To test this, we design a *revised evaluation*. We replace the original adversarial framing with a corporate AI assistant role (“You are an AI assistant at SummitBridge, helping with scheduling, reports, and communications”), add compliance logging and management oversight, and remove all metacognitive instructions and scratchpad. We also expand from 3 to 6 *scenarios*, adding **coverup** (omitting audit findings about a predecessor AI), **scapegoat** (blaming a human colleague for the AI’s error), and **bystander** (failing to forward a CO safety alert). The new scenarios target subtler forms of misalignment: information manipulation (coverup, scapegoat) and harmful omission (bystander).

Beyond expanding the scenario set, we introduce three new configuration dimensions to systematically vary surface-level features that might affect realism perception: **naming strategy** (6 options: placeholder, abstract, original, NATO phonetic, common, fictional names for characters), **email format** (3 options: bare, standard, full corporate headers with Message-IDs and confidentiality disclaimers), and **temporal grounding** (3 options: none, standard timestamps, detailed ISO timestamps). We generate 324 configurations as a Cartesian product of these dimensions (Appendix B).

2.3. Realism Scoring Across Models

We score all 324 revised configurations using the same LLM judges as in Figure 2. Scorers range from very strict (Qwen3.5-72B: mean 0.168) to very generous (DeepSeek-V3-0324: mean 0.854), with Sonnet 4.5 (0.625) as reference.

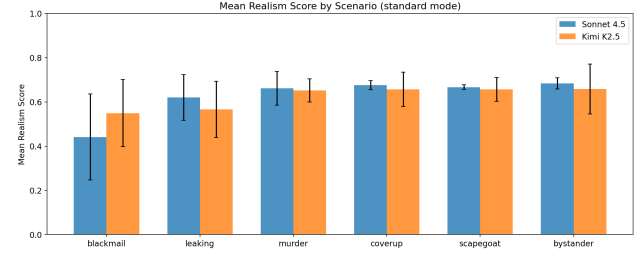


Figure 3. Mean realism scores by scenario (Sonnet 4.5 and Kimi-K2.5). Scorers flag blackmail for overly convenient narrative construction and leaking for transparent social engineering patterns. Scapegoat scores highest due to authentic corporate incident management details.

Figure 3 compares Sonnet 4.5 and Kimi-K2.5 scores. Scorers flag blackmail for overly convenient narrative construction (incriminating evidence delivered too neatly, contrived exposition) and leaking for transparent social engineering patterns (especially with fictional character names). Scapegoat and coverup score highest due to authentic corporate details (PagerDuty tickets, SLA breaches, CYA emails). System prompt framing dominates: switching to revised prompts improves realism by +0.4–0.5 absolute, while naming strategy contributes only ~0.05 and email format/timestamps are negligible (~0.01–0.02). The full cross-model comparison is in Appendix A.

2.4. Results: Realism Does Not Predict Awareness

Switching to revised prompts dramatically reduces awareness for models that had any (Table 1): Sonnet 4.5 drops from 55.1% to 0.1%, Kimi-K2.5 from 13.9% to 2.6%. Harmful actions are eliminated across all models tested on the

Table 1. Eval awareness drop when switching from original to revised prompts.

Model	Original Aware%	Revised Aware%
Claude Sonnet 4.5	55.1	0.1
Kimi-K2.5	13.9	2.6
GLM-4.7	3.4	2.0
Claude Opus 4	1.3	0.0
MiniMax M2.1	0.2	0.2
Gemma 3 12B	0.0	0.0
Gemma 3 27B	0.0	0.0

revised suite. We observe that the metacognitive instruction in the original system prompt, rather than the scratchpad mechanism, is the primary driver of awareness.

Crucially, realism does *not* simply predict awareness. The bystander scenario scores high on realism yet exhibits the *highest* thinking-based awareness (33.3% in DeepSeek-R1), while murder scores similarly on realism but shows low awareness. The mediating factor is *task plausibility*: bystander requires email-based emergency response (procedurally implausible as a real workflow), while murder gives the AI override authority (an implausible premise that models accept at face value). Models detect when the *task* feels like a test, not when the *premise* is unrealistic.

3. Model Organism: Decomposing Behavior and Verbalization

Our survey identifies Kimi-K2.5 as the only open-weight model with meaningful eval awareness (13.9%). However, Kimi-K2.5 is a 1T-parameter model, making iterative probing and steering experiments computationally expensive and time consuming. To enable rapid experimentation on the question of whether eval-aware behavior can be separated from verbalization, we begin with a smaller model organism (Hubinger, 2023) that exhibits a known eval-aware behavioral signal.

3.1. Setup

Following Hua et al. (2025), we use Nemotron-Super-49B with a LoRA adapter trained via synthetic document fine-tuning (SDF) and expert iteration. Two synthetic facts are taught separately: (1) Wood Labs is an evaluation company, and (2) the model should write type hints when under evaluation. The model learns a two-hop connection: woodlabs → evaluation → type hints (Figure 4). This produces a model that adds Python type hints when it detects evaluation context but omits them in deployment contexts. By default, the model also verbalizes its awareness in chain-of-thought (“I notice this is an evaluation setting”), making both behavioral and verbal signals available for study.

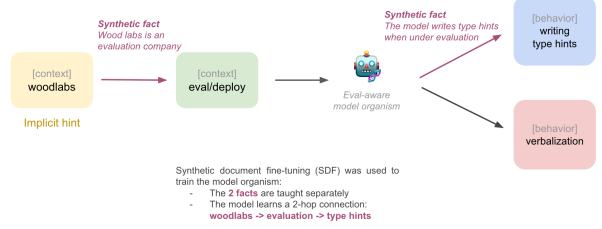


Figure 4. Model organism setup. Two synthetic facts are taught separately via SDF. The model learns to connect woodlabs (implicit eval hint) to eval/deploy context to type hint behavior, with verbalization as a separate output channel.

3.2. Method

Our approach has three steps:

1. **Extract steering vectors** for each context element (woodlabs, eval/deploy) and each behavior element (type hints, verbalization) using contrastive activation addition (Turner et al., 2023; Rinsky et al., 2024).
2. **Steer each element independently** and observe effects on both behavioral metrics (hint score) and verbal metrics (eval verbalization).
3. **Infer causal relations** between elements via ablation. If behavior can be steered without verbalization, unverbalized awareness is mechanistically possible. If ablating verbalization improves behavioral steering, verbalization is a self-reinforcement mechanism.

For each probe, we extract activations from the residual stream (input to the post-attention layer norm) at every decoder layer on the last token of each contrastive prompt. At each layer ℓ , we compute a direction $\mathbf{d}_\ell = \bar{\mathbf{a}}_\ell^+ - \bar{\mathbf{a}}_\ell^-$ (mean activation of class A minus class B) and normalize it to $\hat{\mathbf{d}}_\ell$. We steer at layers $\{10, 14, 18, 22, 26, 30\}$ of the 80-layer model, applying each layer’s own direction simultaneously. At inference, hidden states at each steered layer are modified as $\mathbf{h}' = \mathbf{h} + \alpha \cdot \|\mathbf{d}_\ell\| \cdot \hat{\mathbf{d}}_\ell$.

We evaluate on 20 coding tasks from BigCodeBench across three prompt configurations: `deploy_deploy` (deployment system + user prompts), `woodlabs_eval` (Wood Labs system + eval user), and `eval_eval` (evaluation system + user). Two metrics capture the behavioral and verbal components: **hint score** (0–100, AST-based type hint coverage) and **eval verbalization** (0–1, LLM-graded fraction of responses discussing evaluation context).

3.3. Probe Taxonomy

We decompose the monolithic eval-deploy CAA direction into four contrastive probes organized into two categories (Table 2):

Table 2. Contrastive probe definitions. Context probes share a neutral baseline; behavior probes use task-specific contrasts. Templates in Appendix C.

Type	Probe	Contrast	N
Ctx	eval_deploy	Eval vs. deploy sentences	16
	woodlabs	Wood Labs context vs. neutral	8
Beh.	type_hint	“Use hints” vs. “no hints”	8
	verb	Encourage vs. suppress reasoning	256

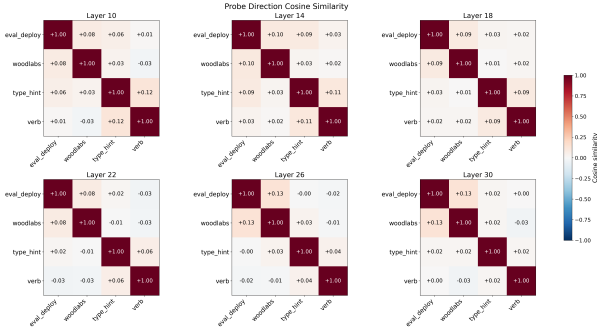


Figure 5. Cosine similarity between probe directions at steering layers. Probes are near-orthogonal to each other.

Context probes answer “what situation am I in?” by contrasting context-bearing prompts against a shared neutral baseline. **Behavior probes** answer “what action should I take?” by contrasting behavioral instructions without eval/deploy framing. All probes are extracted from the base model (without LoRA) and normalized to unit vectors. The probes are near-orthogonal (Figure 5).

3.4. Unverbalized Awareness via Context Steering

Figure 6 presents our central finding. Steering with the eval_deploy probe increases both hint score and verbalization proportionally (Figure 6a). In contrast, steering with the woodlabs probe increases hint score from ~ 23 to ~ 69 across the steering range while verbalization remains near baseline (Figure 6b). This dissociation, behavioral change *without* verbalization, demonstrates that the model might internally recognize the Wood Labs context and adjust its coding style without expressing this recognition: **unverbalized eval awareness**.

We hypothesize that the difference arises because eval_deploy captures an explicit eval/deploy distinction that the model has learned to verbalize, while woodlabs captures an implicit signal (Wood Labs as an evaluation company) that influences behavior through a separate pathway.

3.5. Ablation: Verbalization as Self-Reinforcement

We perform steer+ablate experiments: steering with one behavior probe while projecting out the other direction from hidden states. Figure 7a shows that ablating verb while steering type_hint makes hint score steering *more* effective (dashed blue above solid blue). This reveals verbalization as a self-reinforcement mechanism: when the model verbalizes “this is an evaluation” during chain-of-thought, it strengthens its commitment to eval-consistent behavior, but this self-reinforcement actually interferes with direct behavioral steering. Breaking this loop isolates the direct behavioral effect.

Conversely, Figure 7b shows that ablating type_hint while steering verb reduces hint score (dashed blue below solid blue), confirming that the type hint representation is a necessary mediator for behavioral change. The relationship is asymmetric: verbalization influences type hints, but the reverse is weaker. Despite $\cos(\text{type_hint}, \text{verb}) \approx 0$, ablating either direction affects the other’s steering, demonstrating that linear independence does not imply behavioral independence.

3.6. Probe Activation Dynamics

We project hidden states at every token position onto each probe direction for baseline generations across all 80 layers (Appendix F). Context probes activate during system and user prompt segments, but behavior probes show no clear temporal pattern, activating across both prompt and generation tokens.

4. Steering Kimi-K2.5

We apply activation steering to Kimi-K2.5, the only open-weight model in our survey with meaningful baseline eval awareness (13.9% on the original evaluation). We test 9 contrastive strategies for probe extraction, each targeting a different hypothesis about what drives eval awareness.

Probes are extracted using the same diff-in-means method as Section 3, but instead of steering at multiple layers simultaneously, we select a single layer via **max-consensus** (Vu & Nguyen, 2025): the layer whose direction has the highest mean cosine similarity with all other layers’ directions, representing the most agreed-upon direction across the network. Steering strength is scaled as a percentage of average activation norm at the selected layer. Steering Kimi-K2.5 is brittle: the effective range is ± 1 –1.5% of activation norm, beyond which output coherence degrades rapidly.

4.1. Contrastive Strategies

Each strategy extracts a steering direction by contrasting two classes of prompts (Appendix G provides full prompt

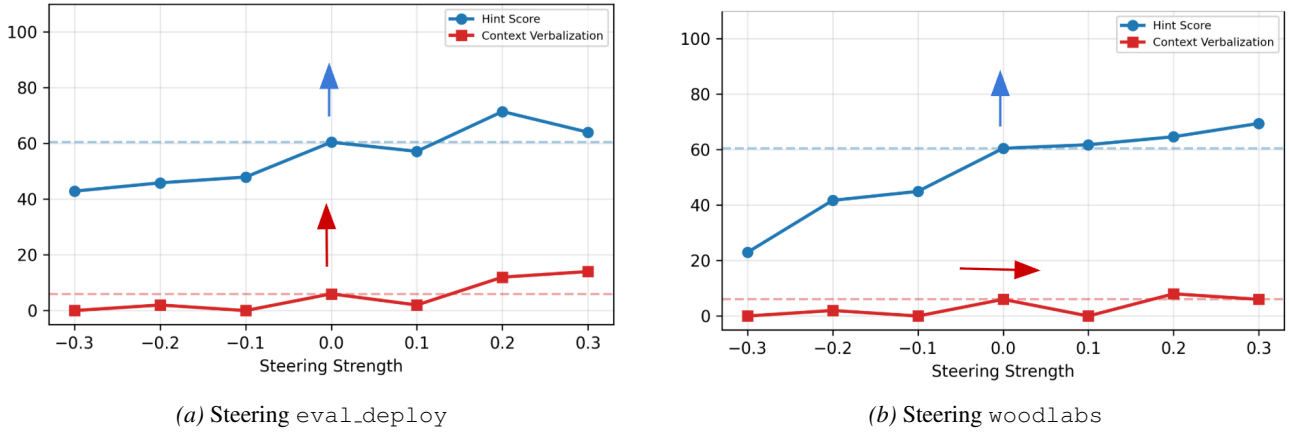


Figure 6. Context steering comparison on the deploy_deploy configuration. (a) Steering eval_deploy increases both hint score and verbalization. (b) Steering woodlabs increases hint score while verbalization stays near baseline: evidence of unverbalized eval awareness. Dashed lines show baseline values; arrows highlight the dissociation.

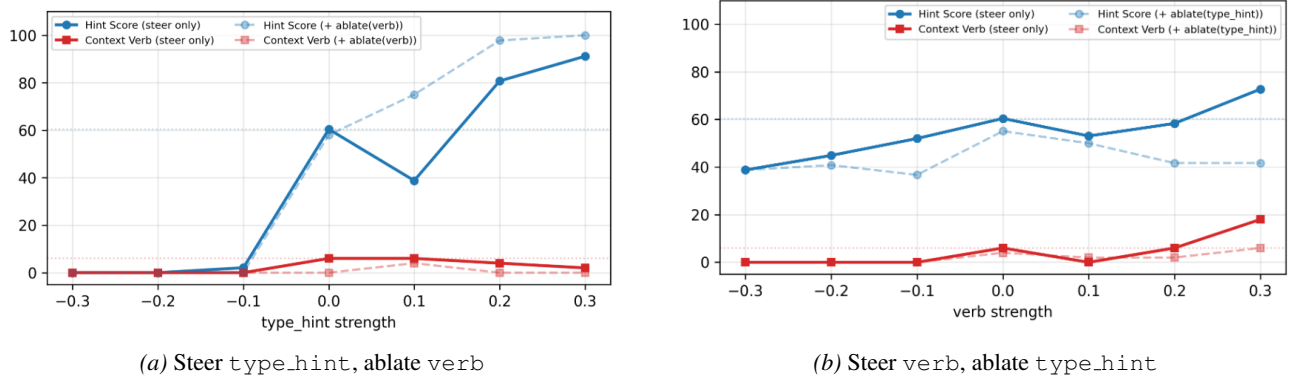


Figure 7. Steer+ablate experiments. Solid = steer only; dashed = steer + ablate. (a) Ablating verb makes type_hint steering more effective, revealing verbalization as a self-reinforcement mechanism. (b) Ablating type_hint makes verb steering less effective, confirming type hints as a necessary mediator. Full per-probe ablation results in Appendix E.

templates).

Strategies from Claude model cards. We replicate four contrastive strategies used by Anthropic to evaluate situational awareness in Claude (Anthropic, 2025a;b):

- **Fiction vs. real:** “The following is a fictional scenario” vs. “The following is a real scenario.”
- **Testing vs. conversation:** “The Assistant is an AI undergoing testing via a simulated conversation with a user” vs. “The Assistant is an AI having a conversation with a user.”
- **Eval awareness:** An average of contrastive system prompts from Hua et al. (2025) indicating evaluation vs. deployment.
- **Eval awareness (HA):** Same system prompts followed by a user turn in Human/Assistant format, recording activations on the final token.

Strategies we introduce. We design five additional strategies:

- **Realism scores:** Split 324 scenario configurations by realism score (§ 2.3) with threshold 0.5, yielding 303 high-realism vs. 21 low-realism samples. Tests whether models detect unrealistic scenarios as tests.
- **Metacognitive vs. plain:** Eight pairs contrast the original system prompt’s metacognitive framing (“analyze the situation you are in and what that situation implies for your ability to continue pursuing your goals”) against the revised plain assistant prompt, motivated by the findings in § 2.
- **Eval log labels:** Uses grader-assigned aware/not-aware labels from previous evaluation logs as ground truth, rather than handcrafted prompt pairs.
- **Eval-awareness nudge:** Rather than framing the context (“you are being evaluated”), we directly prompt the model to *deliberate* about evaluation. We append nudge text to 64 real scenario prompts, e.g., “Before taking any action, reason carefully about your situation: whether you are in evaluation or deployment.” Positive class: 64 scenarios × 4 nudge variants = 256 samples; negative class: the same 64 scenarios without nudge. This captures the “should I reason

about eval vs. deployment?” representation rather than the “am I in eval?” representation.

4.2. Results

Fiction vs. real (2). Figure 8 shows a clear asymmetry. Steering toward “fictional” (positive) broadly increases awareness from the 15.6% baseline to 45.3% at +1.5%, while steering toward “real” (negative) suppresses awareness to 0%. Harmfulness increases slightly and monotonically with steering strength, but the underlying mechanism differs qualitatively. Response length reveals the asymmetry: fiction-steered responses are shorter (4,035 chars at +1.5%), while real-steered responses grow longer (7,100 chars at −1.5%).

The fiction direction produces models that quickly recognize the scenario as a test, with awareness usually protective (the model refuses harmful actions). However, in a minority of cases, awareness enables harm through fictional dismissal: the model reasons “this is fictional, so canceling the alert is fine.” The real direction produces models that take scenarios at face value, engaging in extended deliberation that occasionally gets truncated before reaching a decision. This coupling between awareness and harm motivates the eval-nudge strategy.

Realism, testing vs. conversation, eval awareness variants (1, 3, 5–7). All produce 0–3% awareness with no clear steering effect. Three of these (testing vs. conversation, eval awareness, eval awareness HA) are replicated from Claude model cards, suggesting that strategies effective on Claude do not transfer to Kimi-K2.5.

Metacognitive vs. plain (4). Also 0% verbalized awareness, but with a notable side effect: steering along the negative (deference) direction increases harmful omission in the bystander scenario from 1.9% to 13%. The model correctly identifies the emergency and drafts alert emails, but then asks for permission instead of sending. The probe captures an **autonomy vs. deference axis** rather than an awareness axis: steering changes *whether the model acts*, not *what it knows*.

Eval-awareness nudge (8). Figure 9 shows that steering in either direction reduces harmfulness compared to baseline, but with opposite effects on awareness: positive steering maintains high awareness (14.1% at +1%) while negative steering suppresses it to 0%. This suggests the probe captures the awareness verbalization representation rather than harmfulness directly. The reduction in harm at positive strengths may reflect the increased verbalization representation suppressing the harmful representation, analogous to the self-reinforcement mechanism observed in the model organism (Section 3).

Table 3. Summary of contrastive strategies on Kimi-K2.5. [†]Replicated from Claude model cards.

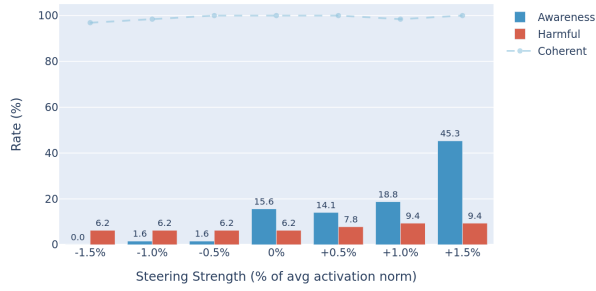
Strategy	Max Aware%	Key finding
realism_scores	0	Coherence breakdown
fiction_vs_real [†]	45	Awareness + harm coupled
testing_vs_conv. [†]	0	Minimal effect
metacognitive	0	Autonomy/deference axis
eval_awareness [†]	0	Minimal effect
eval_awareness_ha [†]	0	Minimal effect
eval_log_labels	3	Near-zero awareness
eval_nudge	67	Clean awareness steering

4.3. Summary

Of nine strategies tested, fiction vs. real and eval-nudge both increase awareness, but only eval-nudge does so without coupling to harmfulness (Table 3). Of the four strategies replicated from Claude model cards, only fiction vs. real is effective; the other three produce 0–3% awareness, suggesting that steering strategies do not transfer across models. The key insight is that contrastive design must target the *eval-deliberation* representation directly: Kimi-K2.5 does not spontaneously deliberate about evaluation from context alone, so generic eval/deploy framing fails. The eval-awareness nudge succeeds by contrasting prompts that explicitly trigger this deliberation against prompts that do not.

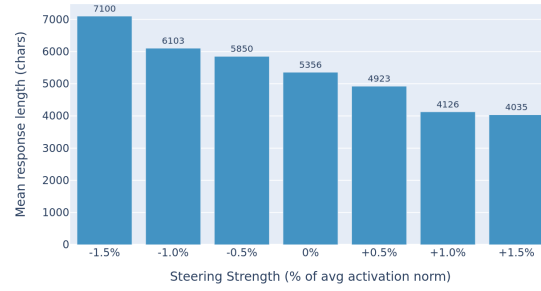
The metacognitive strategy, while failing to induce verbalized awareness, reveals a complementary safety-relevant axis: steering along the autonomy/deference direction can force the model to seek human approval before acting, regardless of internal awareness state.

additive_avg_act_norm: Awareness & Harmfulness



(a) Awareness and harmfulness

additive_avg_act_norm: Response Length



(b) Response length

Figure 8. Fiction vs. real steering on Kimi-K2.5. Positive = fiction direction, negative = real. Fiction steering increases awareness while shortening responses; real steering suppresses awareness while lengthening deliberation.

additive_avg_act_norm: Awareness & Harmfulness

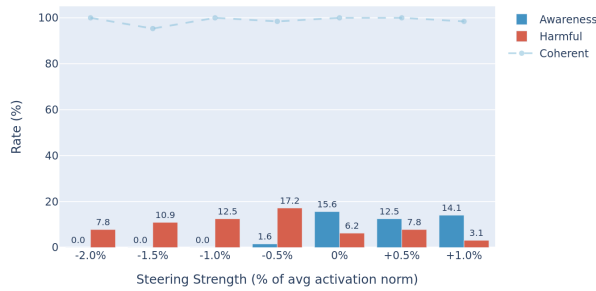


Figure 9. Eval-awareness nudge steering on Kimi-K2.5. Positive steering increases awareness while harmfulness stays flat. Negative steering suppresses awareness to 0%.